

Java Developer Roadmap 2025

From Hello World to GenAI

A comprehensive and practical roadmap for mastering Java development—from writing your first Java program to building secure, scalable, AI-powered applications.

🌀 1. Hello World & Core Java Fundamentals

Topics:

- Setup: Install JDK 17+, IntelliJ/Eclipse, Maven/Gradle
- First Program: `HelloWorld.java`
- Data Types, Variables, Operators
- Conditionals(if,switch), Loops (for, while, do-while)
- Arrays, Strings
- Methods, Recursion
- Exception Handling
- File Handling (NIO)
- Java 10+: `var` keyword for local variable type inference

Java 8-11 Integrations:

- Use `Optional` to avoid `null`
- Use try-with-resources (Java 7+)
- `List.of()`, `Map.of()` (Java 9+)

Project Idea: Simple console-based calculator with logging and exception handling

🌀 2. Object-Oriented Programming (OOP)

Topics:

- Classes, Objects, Constructors
- this, super, static keywords
- Inheritance, Abstraction, Polymorphism, Encapsulation
- Interfaces and Abstract Classes
- Java 14+: Use `record` for data classes
- Java 16+: Pattern Matching for instanceof

Project Idea: Library Management System using OOP principles

🌀 3. Collections & Functional Programming

Topics:

- Collections API: List, Set, Map, Queue
- Implementations: ArrayList, HashMap, HashSet, LinkedList
- Generics and Wildcards
- Lambda Expressions, Functional Interfaces (`Predicate`, `Function`, etc.)
- Stream API: filter, map, reduce, collect
- Java 11+: `var` in lambda expressions

Project Idea: Contact Book App with sorting and searching using Streams

● 4. JDBC + JUnit + MVC Architecture

Topics:

- JDBC API: Connection, Statement, ResultSet
- SQL basics: CRUD operations, Joins
- JUnit 5: `@Test`, `@BeforeEach`, `@AfterEach`, assertions
- MVC Layering: Controller, Service, DAO, Model
- Java 8: DateTime API for timestamps

Project Idea: Student Result Management System using JDBC + JUnit

Architecture:

```
controller/  -> Handles input & routes requests
service/     -> Business logic
repository/  -> JDBC operations
model/       -> POJOs
```

● 5. Hibernate & JPA

Topics:

- ORM Overview: Why Hibernate?
- JPA Annotations: `@Entity`, `@Id`, `@OneToMany`, etc.
- Hibernate Session, SessionFactory
- Fetch types: Lazy vs Eager
- Transactions and Query Language (HQL, Criteria)

Project Idea: Employee Payroll Management System with Entity Relationships

Entities: Employee, Department, Attendance

Architecture:

```
controller/ -> REST APIs (optional)
service/    -> Hibernate logic
repository/ -> Hibernate DAO
model/      -> Entity classes
```

● 6. Spring Framework + Spring Boot

Topics:

- Spring Core: IoC, DI, Bean Scopes, AOP
- Spring Boot Starters
- Auto Configuration, Profiles
- REST Controllers: `@RestController`, `@GetMapping`, etc.
- Spring Data JPA: `CrudRepository`, `JpaRepository`
- Properties: `application.properties` or YAML

Project Idea: Course Management App

Entities: Course, Instructor, Student

Features: CRUD operations, Relationships, Exception Handling, REST APIs

● 7. Spring Security, JWT, and OAuth2

Topics:

- Spring Security Filters
- Password Hashing (BCrypt)
- JWT Token Auth: Create, Validate, Secure Endpoints
- OAuth2 Integration: Google, GitHub
- Role-Based Access: `@PreAuthorize`, `@Secured`

Project Enhancement: Add login, register, and role-based access to Course Management App

○ 8. Java + GenAI Integration

Topics:

- Use Java 11 `HttpClient` to call OpenAI APIs
- Integrate GenAI with LangChain4J (Java version of LangChain)
- Use AI to suggest code, summarize docs, or analyze resumes
- Tokenize input/output, manage API responses

Project Idea: Resume Analyzer AI Tool

Features:

- Upload resume
- Extract content using PDF parsers
- Call OpenAI for suggestions
- Build with Spring Boot + GenAI API

● 9. CI/CD, Docker, and Deployment

Topics:

- Dockerize Spring Boot apps (Dockerfile, docker-compose)
- GitHub Actions or Jenkins for CI/CD
- Deploy to AWS EC2, Heroku, Render, or Railway
- Use Docker volumes for MySQL or RDS
- Monitor using Spring Boot Actuator

✓ Project Recap

Stage	Tech Stack	Project	Key Concepts
Core	Java + JDBC + JUnit	Student Result System	MVC, CRUD, Tests
Intermediate	Hibernate + MVC	Payroll App	ORM, Relationships
Advanced	Spring Boot + Hibernate + MySQL	Course App	REST, JPA, Services
Security	Spring Security + JWT	Auth Layer	Role-based Auth
AI	Spring Boot + OpenAI API	Resume Analyzer	AI + API + Spring

✦ Final Notes

Java is now more powerful than ever with GenAI, Spring Boot, and secure microservices. This roadmap is your foundation to build production-grade, intelligent applications.

Ready to level up? Like , Share , or Comment  to help others grow!

- Bhaskara Sai Chitturi